Do Teams' Passing Network Change Significantly When Winning, Drawing or Losing Matches?

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Introduction

Football matches involve continual interactions between players on the same team, who work together to create attacking situations by carrying or passing the ball. Over a match, a team's passes can therefore be considered as a player passing network (Buldú et al., 2018), where players are represented by nodes and passes between them by directed edges.

Passing networks may be considered statically, representing the cumulative number of passes across a whole match or up to a specific time point, such as half-time. This is useful for evaluating overall team or player performance (Pan et al., 2025), and has also been extended to multiplex networks, where different pass types are represented across separate layers (Beheshtian-Ardakani et al., 2023). Passing network may also be considered as temporal networks (Holme & Saramäki, 2012). In a temporal passing network, directed edges exist at the timestamp at which the pass occurred between player i and player j. While this network can still be observed cumulatively at a given time point, the added temporal dimension allows network dynamics to be explored. For example, Cao (2025) used non-overlapping 2 and 5-minute sliding windows over pass-event data from the 2018 FIFA World Cup Final to examine how graph metrics such as global clustering coefficient and network centralisation changed over the match, and to identify emergent passing "states" through unsupervised clustering.

Temporal passing networks may vary with match status. For instance, a team winning early may press their advantage and increase passing tempo, whereas a team winning late may reduce tempo and circulate possession to control the game. Previous work has shown that match status can influence football behaviours such as possession strategy and passing sequences (Lago, 2009; Paixão et al., 2015). Praça et al. (2019) examined this during the 2018 FIFA World Cup by constructing adjacency matrices for each team's winning, drawing and losing states. They found that density and clustering did not significantly differ between states. However, aggregating passes from each match state into single matrices may obscure within-match variation, since winning periods occur at different moments and under different levels of fatigue, pressure, or tactical adjustment. A rolling-window temporal network approach instead preserves local changes in network structure and allows the same team's passing behaviour to be compared across short periods. This enables analysis of whether match status is associated with changes in network topology and dynamics over time, rather than only in aggregate.

This project therefore aims to explore how teams' temporal passing networks differ when

they are winning, drawing, or losing. Using event data from 326 WSL matches across three seasons, temporal passing networks were constructed with non-overlapping five-minute rolling windows, each labelled with the team's match status at that time. Topological metrics, including global clustering coefficient, density, reciprocity and out-degree centralisation, and temporal metrics, including median inter-event time and burstiness, were then measured across each temporal window rather than in a single aggregate network for each match state.

Methods

Dataset Card

The dataset used in this report corresponds to a set of 652 temporal association football passing networks corresponding to each team's set of completed passes from all 326 WSL matches played across the 2018/19, 2019/20 and 2020/21 seasons.

These networks are derived from datasets from StatsBomb's Open Data repository (StatsBomb, 2026a), retrieved via the `statsbombpy` package (StatsBomb, 2026b). The `events` and `matches` data from each season were imported and preprocessed to combine match data such as kick-off times with events in order to produce complete timestamps for each pass event, necessary to produce temporal edges.

The public data user agreement for the StatsBomb Open Data repository can be found at: https://github.com/statsbomb/open-data/blob/master/LICENSE.pdf

This dataset is intended to support exploration of this report's research question, but could feasibly be used for any research project involving the study temporal passing networks in soccer. This dataset does not constitute a random sample of football matches as such, instead being limited to a handful of season from a particular competition. Inferences made from this dataset may be seen as representative of passing networks for WSL matches during $2018 - 21$, which may not reflect different time periods or competitions.

Dataset Composition

Figure 1 shows the overall degree distribution across all 652 temporal graphs. We can see that $k = 10$ comprises the most frequent average degree count across the graphs. Degrees greater than 10 are present due to substitutions, which shows lower average frequencies reflecting their lower amount of playing time and thus opportunities to pass or be passed to.

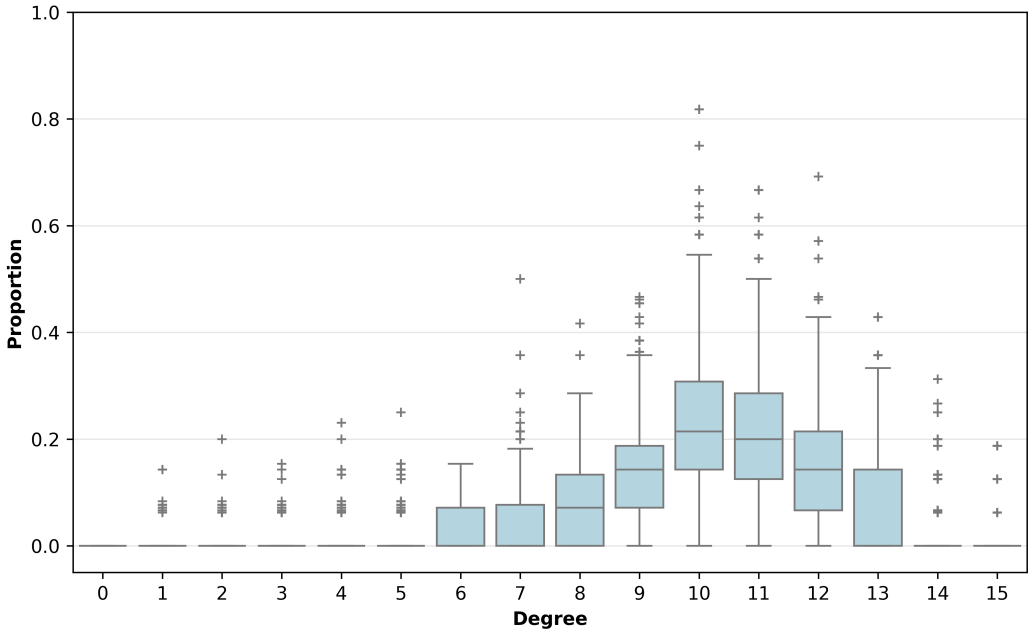


Figure 1. Boxplot for Degree Distribution Across All Graphs

This project uses 6 graph metrics in total. Figure 2 shows the distribution of each of these metrics across each graph. For each graph, each metric is measured over 5-minute non-overlapping rolling windows and then averaged to produce an overall score for the graph.

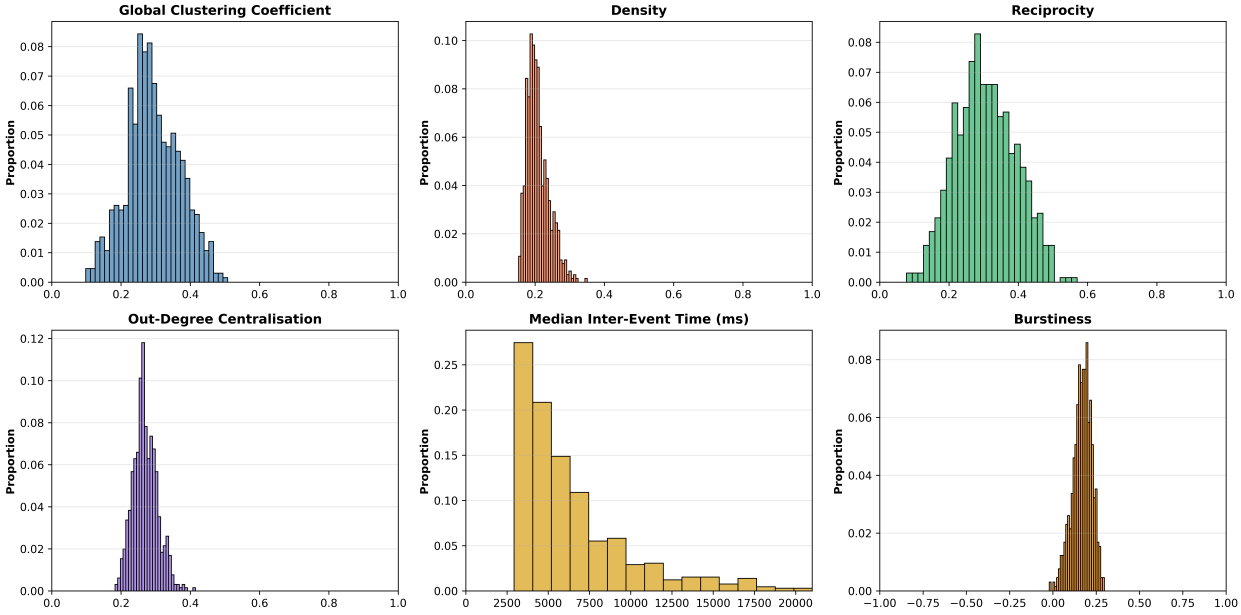


Figure 2. Distributions of Graph Summary Statistics Across All Graphs

Topological Metrics

Clustering Coefficient

One aspect of passing behaviour we can consider is how often trios of players form passing combinations. A common possession playstyle, known as "positional play" (Coaches' Voice, 2021), involves fluidly moving players to create numerical overloads and passing combinations to progress the ball into dangerous attacking positions. This contrasts with "direct play", which avoids extended passing combinations and instead moves the ball quickly into dangerous areas (Tenga et al., 2010). These behaviours may also vary by match status: winning teams may become more clustered earlier in matches as they seek to dominate possession and control the game, whereas teams losing late may become less clustered as they play more directly in pursuit of an equaliser. One graph metric which can measure these variations in passing style is the global clustering coefficient.

$$C = \frac{N_{\text{closed triplets}}}{N_{\text{triplets}}} \tag{1}$$

The global clustering coefficient measures the proportion of closed triplets to total triplets in the network. Figure 3 shows that a closed triplet requires a directed edge between each node in the triplet, whereas a triplet is simply three nodes connected by two directed edges, with edge direction allowed to vary. A higher clustering coefficient therefore corresponds to a greater proportion of passing triangles within a temporal window, potentially indicative of positional play.

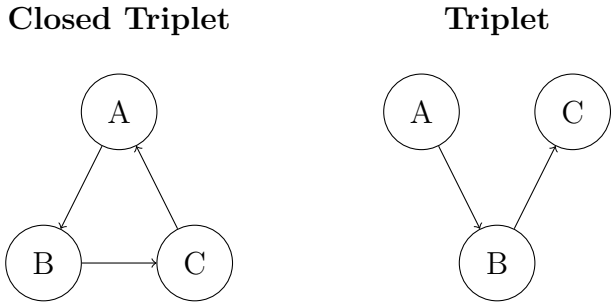


Figure 3. Examples of a Directed Closed Triplet and a Directed Triplet

Density

Extending beyond passing triangles, we can also consider more globally how connected players are via passes across the temporal network. We might expect teams' passing networks to become denser during winning phases as they utilise more players to control possession, whereas losing teams may exhibit lower density as they struggle to maintain possession.

For a given temporal window, there exists a maximum number of unique links between pairs of players. If n players are involved, this maximum is simply $n \times (n-1)$. We can therefore consider density as the proportion of actual observed links L to this maximum.

$$D = \frac{L}{n(n-1)} \tag{2}$$

Reciprocity

Directness in passing behaviour may also be captured by considering how often a pass from player $i \to j$ is reciprocated by $j \to i$ within a temporal window. More direct passing behaviour, potentially during drawing phases late in a match where both teams attempt to score a winning goal, would exhibit fewer reciprocal passes, as the aim is to progress the ball quickly into attacking situations rather than cycle possession through the team. Reciprocity therefore considers the ratio of reciprocated edges to the total number of edges in a temporal window.

$$r = \frac{E_{i \to j, j \to i}}{E_{\text{total}}} \tag{3}$$

Out Degree Centralisation

The previous metrics help to characterise passing styles. We might also consider the role of playmakers across different match statuses. If one or a few players are responsible for a disproportionate number of passes, this may indicate playmakers influencing passing behaviour. A winning team may conceivably become less reliant on a single playmaker as they circulate the ball among more players, whereas a drawing team may funnel possession through specific players who are effective at creating chances.

Out degree refers to the number of outgoing directed edges from a node, or how many passes a player made. Since playmakers tend to make more passes than their team's average, they would usually have a higher out degree. Considering the out degree of every node within a temporal window, we can compare each value with the highest out degree in the network. Summing these differences and normalising by the maximum possible sum gives out degree centralisation. A higher value therefore suggests greater reliance on playmakers during a temporal window.

$$C_{\text{out}} = \frac{\sum_{i=1}^{n} (k_{\text{out}}^{\max} - k_{\text{out}}^{i})}{(n-1)(n-2)} \tag{4}$$

Temporal Metrics

Whilst topological metrics evaluate passing behaviour through the overall structure of the network, temporal metrics measure dynamics in how the network evolves over time.

Median Inter-Event Time

Passing behaviour is often characterised by tempo. A faster passing tempo may indicate a team chasing a winning or equalising goal, whereas a lower tempo may reflect a more

controlled approach by a team already in the lead. To explore tempo, we can arrange the time intervals $\Delta(t_{i-1}, t_i)$ between n passes for a given temporal window in ascending order.

$$\Delta(t_2, t_1) \leq ... \leq \Delta(t_{i-1}, t_i) \leq ... \leq \Delta(t_{n-1}, t_n) \tag{5}$$

The median inter-event time takes the median of this ordered interval sequence, providing an indication of passing tempo whilst avoiding the influence of outliers, such as one long interval in an otherwise fast-tempo passing network. A higher median inter-event time therefore suggests a lower passing tempo.

Burstiness

Median inter-event time gives a measure of overall passing tempo. However, passes within a window may occur irregularly. A more regular tempo may indicate a controlled possession approach by a winning team, whereas greater irregularity may reflect a drawing team battling for possession amid high turnover.

Given a set of time intervals from temporal edges within a temporal window, burstiness can be calculated using the mean μ and standard deviation σ of these intervals. The ratio of their difference to their sum provides a burstiness score. A higher burstiness value suggests greater irregularity in passing intervals.

$$B = \frac{\sigma - \mu}{\sigma + \mu} \tag{6}$$

Analysis Outline

This project will begin by measuring each graph metric across 5-minute non-overlapping rolling windows of each team's temporal passing network for each match. At each interval, each metric is measured alongside whether the team was winning, drawing or losing. Goal events complicate this approach, as they may fall inside a window, meaning the window contains two match states. To avoid this overlap, windows containing a goal event are excluded. The median of these window measurements is then calculated. The median is used because the metric distributions in Figure 2 are not normally distributed, making parametric significance tests inappropriate.

To determine whether median graph values differ significantly across winning, drawing and losing states, ranked values across match status groups are compared using a Kruskal-Wallis test. Where this test is significant, a post-hoc Dunn's test is used to compare each match-status pair across each metric.

To assess effect size, Cliff's delta (Meissel & Yao, 2024) is used to compare two groups by identifying which group's values tend to be higher. A value of 1 indicates that every value in group A is higher than every value in group B, -1 indicates the opposite pattern, and 0 indicates identical distributions.

$$\delta = \frac{\left|\left(x_i^{(1)} > x_i^{(2)}\right) - \left(x_i^{(1)} < x_i^{(2)}\right)\right|}{n_1 n_2} \tag{7}$$

Multiple pairwise significance tests increase the probability of observing at least one false positive. To account for this, Bonferroni-corrected p-values are reported and used for interpretation. Analyses use a significance threshold of $\alpha = 0.05$, with corrected p-values calculated by multiplying the raw p-value by the number of tests, m. For the Kruskal-Wallis tests, $m = 6$, as one test is performed for each metric. For Dunn's tests, three pairwise comparisons are performed for each metric, so $m = 3$ within each set of tests.

$$p_{\text{Bonferroni}} = p_{raw} \times m \tag{8}$$

One potential confound is pass count, as a higher volume of passes creates more opportunities for passing combinations, potentially increasing metrics such as global clustering coefficient and density. Winning states are likely to contain more passes than losing states because winning teams generally have higher possession. To examine the importance of match status while controlling for pass count, an OLS linear regression model is fitted to each metric individually, using `winning vs losing`, `drawing vs losing` and `pass_count` as predictors. Although significant differences may be observed in the earlier tests, these regressions indicate whether observed differences are largely explained by different pass counts between match states. Again, p-values are Bonferroni corrected.

Results

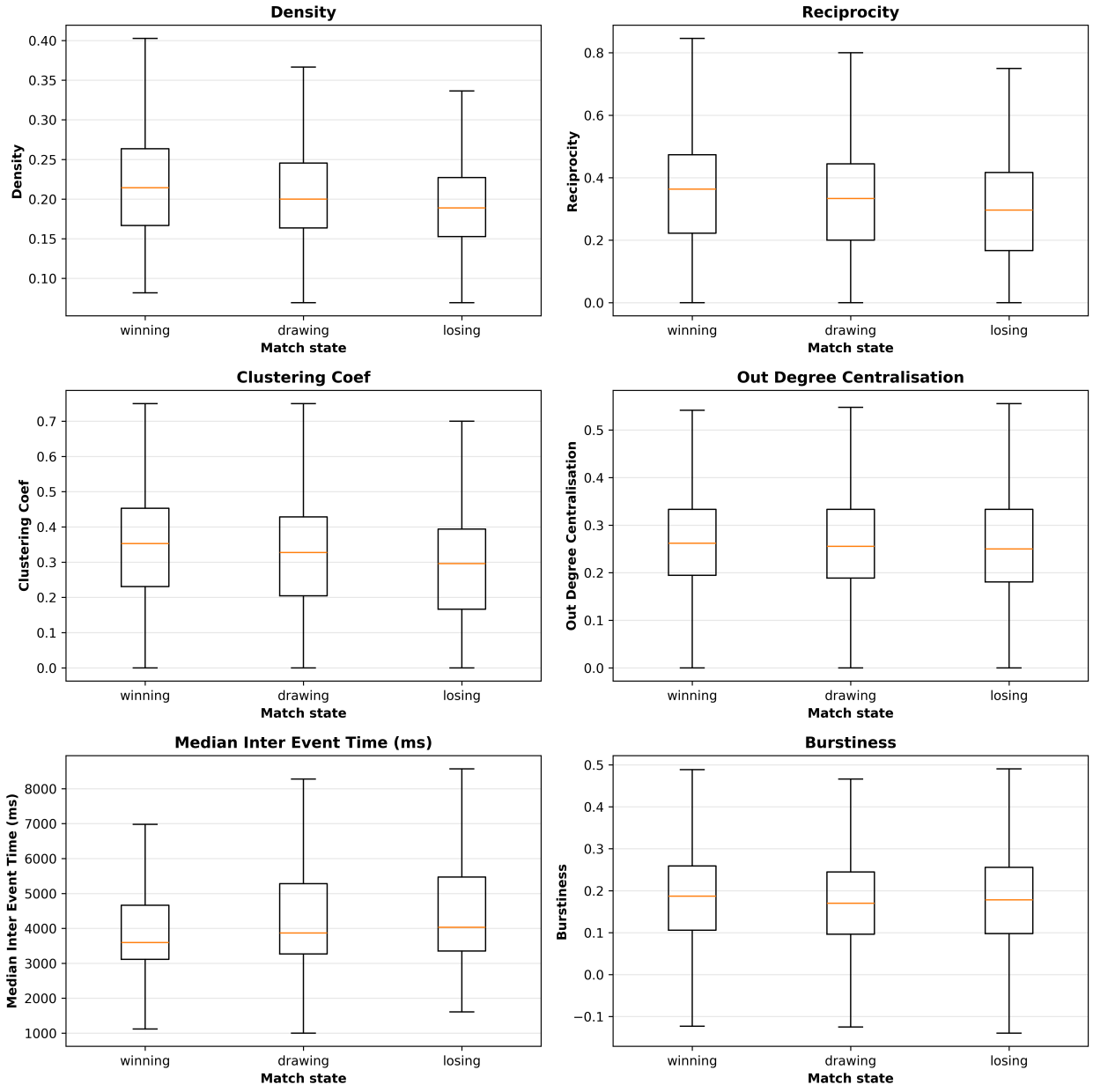


Figure 4. Graph Metrics Boxplots by Match Status

Figure 4 shows boxplots for each match status across each graph metrics being measured. Density, reciprocity and global clustering coefficient subplots show winning match statuses are slightly higher on average, followed by drawing and then losing statuses. Out degree centralisation and burstiness do not appear to vary across match status, whilst median inter-event time shows losing statuses to be slightly higher than drawing which is also higher than winning. It is worth noting that median differences are quite small with a similar range across

match statuses for each graph metric, possibly suggesting that different passing behaviours (e.g. positional vs direct play) are being aggregated across match statuses.

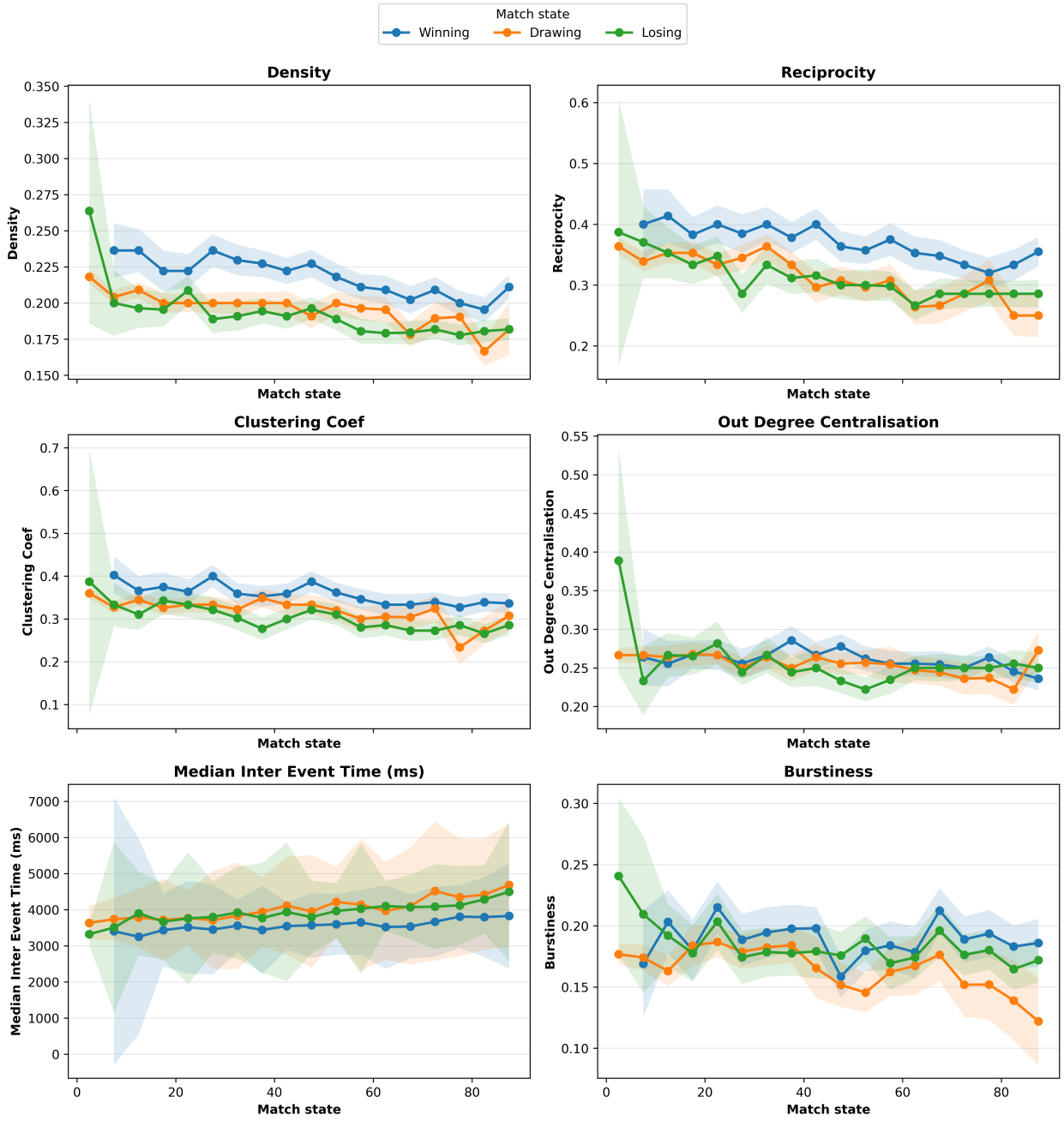


Figure 5. Graph Metrics Over Time by Match Status

Figure 5 shows each graph metric plotted over 90 minutes in each match, split by match status. Density, reciprocity and global clustering coefficient appear to decrease slightly over the match across all statuses, whilst winning statuses remain consistently higher than drawing and losing states. This suggests that, when winning, teams involve more players in possession, are slightly less direct and involve more passing triangles compared to drawing

and losing statuses. In line with Figure 4, out-degree centralisation and burstiness do not appear to differ across match statuses over the match. Interestingly, burstiness appears to reduce in the last 15 minutes for drawing teams, suggesting passing tempo becomes slightly more regular towards the end. Median inter-event time generally increases over the match for all match states, possibly due to player fatigue. Although variance within each status appears quite large, winning statuses appear to have consistently lower median inter-event times than drawing and losing statuses.

Metric	Winning median	Drawing median	Losing median	Kruskal statistic	Raw p-value	Corrected p-value
burstiness	0.178	0.176	0.173	2.556	0.279	1.000
clustering coef	0.321	0.317	0.276	44.723	0.000	0.000
density	0.212	0.206	0.190	59.347	0.000	0.000
median inter event time	4260.500	4296.536	4916.500	21.019	0.000	0.000
out degree centralisation	0.269	0.271	0.262	7.260	0.027	0.159
reciprocity	0.331	0.332	0.291	34.608	0.000	0.000

Table 1. Kruskal-Wallis Significance Test Results

Table 1 shows results from the Kruskal-Wallis tests. Reflecting the patterns observed in Figure 4 and Figure 5, median burstiness ($H(2) = 2.556, p = 1$) and median out degree centralisation ($H(2) = 7.260, p = 0.159$) do not significantly differ across match statuses.

Metric	State A	State B	A median	B median	Corrected p-value	Cliff's delta
clustering coef	drawing	losing	0.317	0.276	0.000	0.247
clustering coef	winning	drawing	0.321	0.317	1.000	0.012
clustering coef	winning	losing	0.321	0.276	0.000	0.251
density	drawing	losing	0.206	0.190	0.000	0.267
density	winning	drawing	0.212	0.206	0.476	0.061
density	winning	losing	0.212	0.190	0.000	0.308
median inter event time	drawing	losing	4296.536	4916.500	0.000	-0.166
median inter event time	winning	drawing	4260.500	4296.536	1.000	-0.025
median inter event time	winning	losing	4260.500	4916.500	0.000	-0.176
reciprocity	drawing	losing	0.332	0.291	0.000	0.212
reciprocity	winning	drawing	0.331	0.332	1.000	0.029
reciprocity	winning	losing	0.331	0.291	0.000	0.228

Table 2. Pairwise Significance Test Results

Table 2 shows results from post-hoc Dunn's tests for each metric which exhibited significant Kruskal-Wallis results. For each metric, winning and drawing statuses significantly differ from losing statuses, whilst there were no observed significant differences between winning and drawing statuses.

For the global clustering coefficient, we observe an effect size of $\delta = 0.251$ and $\delta = 0.247$, for winning vs losing and winning vs drawing, respectively. Meissel & Yao (2024) suggest these generally correspond to a "small" effect size. Density again reveals winning and drawing statuses show small effect sizes of $\delta = 0.308$ and $\delta = 0.267$, respectively. Median inter-event time shows smaller effect sizes ($\delta = -0.176$ and $\delta = -0.166$, respectively), which are both negative and in line with the pattern observed in Figure 4 and Figure 5. Lastly, reciprocity again shows small effect sizes of $\delta = 0.228$ and $\delta = 0.212$, respectively.

From these results, we generally observe that losing statuses are distinct from other statuses across each metric, showing overall low global clustering coefficient, density, reciprocity and higher median inter-event time. Losing states are therefore characterised by significantly less usage of passing combinations, incorporation of the entire team during possession, more direct passing and a lower passing tempo. The non-significant finding between winning and drawing statuses is somewhat in line with findings from Praça et al. (2019), suggesting that at least when considered in aggregate, winning and drawing statuses do not significantly differ in terms of these 6 metrics.

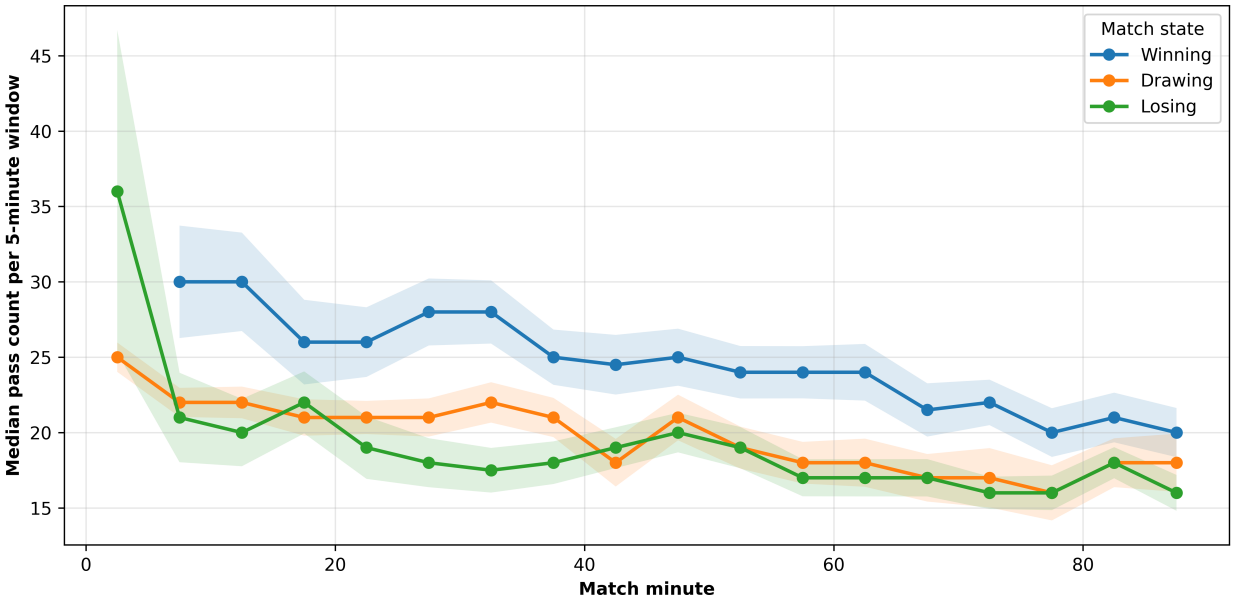


Figure 6. Pass Count Over Time by Match Status

Figure 6 shows how pass counts vary between match statuses over the course of the match. We can see that winning statuses have consistently have higher pass counts during matches, with little difference between drawing and losing apart from the first 5 minutes where losing statuses have much higher pass counts.

Metric	Winning median	Drawing median	Losing median	Kruskal statistic	Raw p-value	Corrected p-value
pass_count	23.000	22.000	18.417	56.988	0.000	0.000

Table 3. Pass Count Kruskal-Wallis Test Results

Table 3 reveals at least 1 match status median pass count significantly differs from the rest.

Metric	State A	State B	A median	B median	Corrected p-value	Cliff's delta
pass_count	drawing	losing	22.000	18.417	0.000	0.285
pass_count	winning	drawing	23.000	22.000	1.000	0.013
pass_count	winning	losing	23.000	18.417	0.000	0.273

Table 4. Pass Count Dunn's Test Results

Post-hoc Dunn's tests shown in Table 4 reveal that, as with the graph metrics, winning and drawing status medians significantly differ from losing statuses, whereas winning and

drawing do not. This finding is somewhat contrasted by Figure 6, where it would seem drawing and losing pass counts are fairly similar across the match, whilst each are fairly lower than winning statuses.

Nonetheless, it is evident that significant differences in pass counts between match statuses are present. This introduced a potential confound for the observed differences between graph metrics.

Metric	Predictor	Coefficient	Std error	t-value	Raw p-value	Corrected p-value	R2
clustering coef	Drawing vs losing	0.001	0.004	0.269	0.788	1.000	0.644
clustering coef	Pass count	0.083	0.002	37.788	0.000	0.000	0.644
clustering coef	Winning vs losing	-0.003	0.005	-0.499	0.618	1.000	0.644
density	Drawing vs losing	0.001	0.002	0.566	0.571	1.000	0.732
density	Pass count	0.034	0.001	43.885	0.000	0.000	0.732
density	Winning vs losing	0.004	0.002	2.293	0.022	0.262	0.732
median inter event time	Drawing vs losing	834.684	331.378	2.519	0.012	0.141	0.287
median inter event time	Pass count	-2896.032	196.072	-14.770	0.000	0.000	0.287
median inter event time	Winning vs losing	1198.450	398.904	3.004	0.003	0.032	0.287
reciprocity	Drawing vs losing	-0.002	0.005	-0.485	0.627	1.000	0.582
reciprocity	Pass count	0.085	0.002	35.292	0.000	0.000	0.582
reciprocity	Winning vs losing	-0.000	0.006	-0.082	0.934	1.000	0.582

Table 5. Regression Results

Table 5 shows results from the regression models constructed to predict each metric using match status and pass count as predictor variables, providing some insight to the relationship between each metric and match status whilst accounting for pass count. For the match status predictor, the losing status is used as the reference class as winning vs drawing median differences are already known to be non-significant.

The regression results show that pass count is a significant predictor in each model, with

each match status predictor value being non-significant, except from winning vs losing for median inter-event time. This model provides an interesting finding, suggesting that once pass count is controlled for, winning match statuses have a higher median inter-event time than losing statuses. This seems to contradict the findings from Figure 5 and Table 2, which show that winning statuses show significantly lower median inter-event times.

Conclusion

This report investigated whether teams' passing networks change significantly when winning, drawing or losing during matches. Temporal passing networks were constructed for each team in 326 WSL matches across 3 seasons. Using non-overlapping 5-minute windows, 6 graph metrics measured the topological and temporal behaviour of each window across 3 match status categories, determined by the scoreline during that window. Values were aggregated across matches to produce median values for each temporal window, then aggregated again to provide a single metric value for each match status within each match. Match statuses were compared using non-parametric significance testing. Given significant differences in pass count, which introduced a confound, regression models were constructed for each metric using match status and pass count as predictors.

Considering the significance tests and regression analysis together, we may conclude that although significant differences between winning vs losing and drawing vs losing statuses were observed for global clustering coefficient, density, reciprocity and median inter-event time, these metrics are largely explained by variation in pass count rather than genuine match-status effects. One result contradicted this pattern: winning vs losing remained a significant predictor of median inter-event time after controlling for pass count. However, this reversed the initial significance-test finding, where winning statuses exhibited lower median inter-event times than losing statuses. Controlling for pass count instead suggested that winning statuses had higher median inter-event times. Overall, much of the variation in passing network behaviour between match statuses appears driven by pass volume.

One limitation of this report is the use of 5-minute rolling windows to produce temporal networks. Within a 5-minute period, possession may change hands multiple times and may not be distributed evenly between winning, drawing and losing teams. This is reflected in the regression findings, which suggest that much variation in passing behaviour is driven by pass volume. An alternative approach would be to segment passing networks by possession, demarked by continuous periods where a team possesses the ball, rather than by fixed-time windows. Whilst fixed-time windows allow simpler comparison, observing passing behaviour within possessions may better account for variation in pass counts and reveal genuine behavioural differences. This windowed approach also required windows involving goals to be excluded, preventing passing behaviour immediately around goals from being observed, despite these periods potentially being interesting and behaviourally distinct. Future work may therefore benefit from moving from fixed windows to possession sequences.

Another limitation is that aggregating only by match status may not provide sufficient resolution to observe meaningful variation in passing behaviour. There is likely substantial

variation between teams within a single match status. For instance, one winning team may play a positional style whilst another adopts a counter-attacking style, with each exhibiting large differences in passing behaviour. Aggregating across tactical styles prevents this from being explored. Future work may seek to identify passing behaviours characteristic of different styles of play, classify passing networks by style, and then observe variation between match statuses.

To conclude, the findings suggest that variation in passing behaviour between temporal networks from different match statuses is largely due to variation in pass count.

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